Exercise Sheet 1 – Solution

Even though this PVK is divided into sections and some content has not yet been covered, the entire material covered in the semester and in the lecture may and should be explicitly used in the series.

Exercise 1
Show that a sequence is convergent if and only if it is a Cauchy sequence.

Solution:

$\Longrightarrow$ Let the sequence $(x_n)_{n\in\mathbb{N}}$ converge to a limit A. Fix $\varepsilon > 0$, then there exists an $N \in \mathbb{N}$ such that $|x_n - A| < \frac{\varepsilon}{2}$ for all $n \geq N$. Let x_n and x_m be terms such that $m, n \geq N$.

$$|x_n - x_m| = |x_n - A - x_m + A| \leq |x_n - A| + |A - x_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \qquad \forall n, m \geq N$$

Thus, $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence.

$\Longleftarrow$ Let $(x)_n$ be a Cauchy sequence. So there exists an N_0 such that $|x_n - x_m| < \frac{\varepsilon}{2}\ \forall n, m \geq N_0$.

We already know that every Cauchy sequence is bounded. Therefore, by the Bolzano-Weierstrass theorem, there exists a convergent subsequence $(x)_{n_k}$. The subsequence converges to A, so there exists an N_1 such that $|x_{n_k} - A| < \frac{\varepsilon}{2}\ \forall k \geq N_1$.

Choosing $N = \max\{N_0, N_1\}$ ensures that both inequalities hold. Because the index of the subsequence n_k is monotonically increasing, we have $n_N \geq N$. Therefore:

$$|x_n - A| \leq |x_n - x_{n_N}| + |x_{n_N} - A| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \qquad \forall n \geq N$$

Exercise 2
Compute the following limits.
Hint: You can use Bernoulli-L'Hospital and Taylor-Approximations.

Solution:

1.
$$\lim_{x\to 1} \frac{\sqrt{x} - 1}{x - 1}$$

We calculate using Bernoulli-L'Hospital:

$$\lim_{x\to 1} \frac{\sqrt{x} - 1}{x - 1} = \lim_{x\to 1} \frac{\frac{1}{2\sqrt{x}}}{1} = \frac{1}{2}.$$

2.
$$\lim_{x\to 0} \frac{\tan(x) - \sin(x)}{x^3}$$

We could use the Taylor Approximations and conclude:

$$\lim_{x\to 0} \frac{\tan(x) - \sin(x)}{x^3} = \lim_{x\to 0} \frac{x + \frac{x^3}{3} + o(x^3) - x + \frac{x^3}{6} + o(x^3)}{x^3} = \frac{1}{3} + \frac{1}{6} + \lim_{x\to 0} o(x^3) = \frac{1}{2}.$$

or we can calculate using Bernoulli-L'Hospital:

$$\lim_{x\to 0}\frac{\tan(x)-\sin(x)}{x^3}=\lim_{x\to 0}\frac{\frac{\sin x}{\cos x}-\sin x}{x^3}=\lim_{x\to 0}\frac{\sin x\cdot\left(\frac{1}{\cos x}-1\right)}{x^3}$$
$$=\lim_{x\to 0}\frac{\sin x}{x}\cdot\frac{1}{\cos x}\cdot\frac{1-\cos x}{x^2}=\lim_{x\to 0}\frac{1-\cos x}{x^2}=\lim_{x\to 0}\frac{\sin x}{2x}=\frac{1}{2}.$$

3.
$$\lim_{x\to 0^+}\frac{e^{-\frac{1}{x}}}{x}$$

We calculate using Bernoulli-L'Hospital:

$$\lim_{x\to 0^+}\frac{e^{-\frac{1}{x}}}{x}=\lim_{x\to 0^+}\frac{\frac{1}{x}}{e^{\frac{1}{x}}}=\lim_{x\to 0^+}\frac{-\frac{1}{x^2}}{-\frac{1}{x^2}e^{\frac{1}{x}}}=\lim_{x\to 0^+}e^{-\frac{1}{x}}=0.$$

We could also make the substitution: $y=1/x$ and calculate the limit as $y\to+\infty$. It then follows directly that

$$\lim_{x\to 0^+}\frac{e^{-\frac{1}{x}}}{x}=\lim_{y\to\infty}ye^{-y}=0.$$

Note that this is rigoros, since $1/x$ is bijective on $\mathbb{R}^+$.

Exercise 3 (FS2024, A. FIGALLI)
Calculate the following Limit.

$$\lim_{x\to 0}\frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x}.$$

Solution:

1. Solution using Taylor-Approximation:

$$\lim_{x\to 0}\frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x}=\lim_{x\to 0}\frac{x+\left(1+o(x^2)\right)\left(3x+o(x^2)\right)}{x^3+o(x^3)+3x+o(x^2)-x}$$
$$=\lim_{x\to 0}\frac{x+3x+o(x^2)}{2x+o(x^2)}=\lim_{x\to 0}\frac{4x}{2x}+\lim_{x\to 0}o(x^2)=2.$$

2. Solution using Bernoulli-L'Hospital:

$$\lim_{x\to 0}\frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x}=\lim_{x\to 0}\frac{1-\sin(x)\sin(3x)+3\cos(x)\cos(3x)}{3\cos(x)\sin^2(x)+3\cos(x)-1}=\frac{1+3}{3-1}=2.$$

Exercise 4 (HS2022, G. FELDER)
Calculate the following Limit.

$$\lim_{x\to\pi}\frac{\sin^2(x)}{\cos(x)+1}.$$

Solution: We calculate using Bernoulli-L'Hospital:

$$\lim_{x\to\pi}\frac{\sin^2(x)}{\cos(x)+1}=\lim_{x\to\pi}\frac{2\cos(x)\sin(x)}{-\sin(x)}=-2\cos(\pi)=2.$$

Exercise 5
Let $D \subset \mathbb{R}$ be a set, let $f : D \to \mathbb{R}$ be a function, and let $\bar{x} \in D$. Show that the function f is continuous at $\bar{x}$ if and only if, for all sequences $(x_n)_n \subset D$ converging to $\bar{x}$, $f((x_n)_n)$ converges to $f(\bar{x})$.

Solution:

$\Longrightarrow$ Let $(x_n) \to x_0$ be a sequence in D and f a continuous function. Fix any $\varepsilon > 0$; according to the continuity of f, there exists a corresponding $\delta > 0$. Since our sequence (x_n) converges to x_0, by definition there exists an $N \in \mathbb{N}$ such that $|x_n - x_0| < \delta \ \forall n \geq N$. Thus, by the continuity of f, we have $|f(x_n) - f(x_0)| < \varepsilon \ \forall n \geq N$. Therefore,

$$\lim_{n\to\infty} f(x_n) = f(x_0)$$

and thus f is sequentially continuous.

$\Longleftarrow$ We show this implication by contraposition. Suppose that f is not continuous. Then there exists an $\varepsilon > 0$ such that for all $\delta > 0$, $|x_n - x_0| < \delta$, but $|f(x_n) - f(x_0)| \geq \varepsilon$. If we choose $\delta = 2^{-n}$, then $x_n \to x_0$ converges in D, but our $f(x_n)$ does not necessarily converge to $f(x_0)$. Thus, f is not sequentially continuous.

Exercise 6
Show that a Lipschitz continuous function f is also uniformly continuous. Give an example of an uniformly continuous function that is not Lipschitz continuous.

Solution: We fix an $\varepsilon > 0$. Let us choose our $\delta = \frac{\varepsilon}{L}$, where L is the Lipschitz constant. We have $|x - \bar{x}| < \delta$ and we can estimate

$$|f(x) - f(\bar{x})| \leq L|x - \bar{x}| < L\delta = L \cdot \frac{\varepsilon}{L} = \varepsilon.$$

We see that δ does not depend on $\bar{x}$! Because $\bar{x}$ was chosen arbitrarily in our domain and $\varepsilon > 0$, f is uniformly continuous.

The function $f(x) = \sqrt{x}$ is uniformly continuous on $[0, 1]$ (continuous and $[0, 1]$ compact), but not Lipschitz continuous, since the derivative $\frac{1}{2\sqrt{x}}$ is unbounded at 0.

Exercise 7
Briefly explain why a uniformly continuous function is always continuous, but a continuous function is not always uniformly continuous.
Note: This does not need to be very rigoros, but a recap of the definitions.

Solution: A uniformly continuous function is always continuous, since uniform continuity is a more specific case in which δ does not depend on the point.
However, a continuous function is not always uniformly continuous, since a priori the δ may depend on our location. This does not interfere with continuity, since a separate δ can be chosen for each location in the domain. In the case of uniform continuity, however, a δ independent of the location is sought, which does not necessarily exist for every continuous function.

Exercise 8 (HS2022, G. FELDER) (a) Let $(f_n)_n$ be a sequence of continuous functions on $D \subset \mathbb{R}$ that converges uniformly to a function f. Show that f is also continuous.

(b) Give an example of a sequence of continuous functions that converge pointwise to a non-continuous function.

Solution:

(a) We fix an $\varepsilon > 0$ and an $\bar{x} \in D$. Since (f_n) converges uniformly to f, we can find an $N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \frac{\varepsilon}{3}$. Furthermore, due to the continuity of f_n, for all $|x - \bar{x}| < \delta$, we have that $|f_n(x) - f_n(\bar{x})| < \frac{\varepsilon}{3}$. Now we estimate

$$|f(x)-f(\bar{x})| \leq |f(x)-f_n(x)|+|f_n(x)-f(\bar{x})| \leq |f(x)-f_n(x)|+|f_n(x)-f_n(\bar{x})|+|f_n(\bar{x})-f(\bar{x})|$$

Thus, for all $|x - \bar{x}| < \delta$, we have $|f(x) - f(\bar{x})| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$. Since ε and $\bar{x} \in D$ were chosen arbitrarily, $f(x)$ is continuous on D.

(b) For example, $f_n(x) = x^n$ on the interval $[0, 1]$.

Exercise 9
We look at the sequence of functions f_n for $n \geq 1$ defined as follows

$$f_n(x) := \frac{x}{1 + nx^2}.$$

(A) Are the functions f_n uniformly continuous? Justify your answer!

(B) Determine the pointwise limit of this sequence of functions.

(C) Do the functions f_n converge uniformly? Justify your answer!

Solution:

(A) For $n \geq 1$ we have, that each f_n is a composition of C^1 functions. Therefore each f_n is C^1 on $\mathbb{R}$. We can therefore analyse wether their derivative is bounded, to obtain Lipschitz continuity.

$$f_n'(x) = \frac{1 - nx^2}{(1 + nx^2)^2}$$

As $x \to \pm\infty$ the functions $f_n' \to 0$. We can also directly see that

$$|f_n'(x)| = \frac{|1 - nx^2|}{|1 + nx^2|^2} \leq \frac{|1 - nx^2|}{|1 - nx^2|^2} = \frac{1}{|1 - nx^2|} \leq 1.$$

By the mean value theorem, each f_n is therefore 1-Lipschitz continuous and therefore uniformly continuous on $\mathbb{R}$.

(B) For $x = 0$ we have $f_n(0) = 0$ and therefore $\lim f_n(0) = 0$. Now fix any $x \neq 0$. We calculate

$$\lim_{n\to\infty} \frac{x}{1 + nx^2} \to 0.$$

The pointwise limit is therefore the zero function $f \equiv 0$.

(C) For uniform convergence we have to find wether

$$\lim_{n\to\infty} \sup_{x\in\mathbb{R}} |f_n(x)| \to 0.$$

We first find the sup of $f_n(x)$ depending on $n \geq 1$. We know from (A) that all f_n vanish at $\pm\infty$. We therefore find the local extrema by setting f_n' to zero. This yields

$$1 = nx^2 \iff x = \frac{1}{\pm\sqrt{n}}.$$

Therefore the global extrema are

$$\left| f\left(\frac{1}{\pm\sqrt{n}}\right)\right| = \frac{\frac{1}{\sqrt{n}}}{|1 + n(\frac{1}{\pm\sqrt{n}})^2|} = \frac{1}{\sqrt{n}|1+1|} = \frac{1}{2\sqrt{n}}.$$

We observe that as $n \to \infty$ the maximal value of $|f_n| \to 0$, therefore we find:

$$\lim_{n\to\infty} \sup_{x\in\mathbb{R}} |f_n(x)| = \lim_{n\to\infty} \frac{1}{2\sqrt{n}} = 0.$$

Thus $f_n \to 0$ as $n \to \infty$ uniformly.

Exercise 10
Let $f : [a, b] \to \mathbb{R}$ be a continuous function with $f(a) \leq f(b)$. Show that for every real number c with $f(a) \leq c \leq f(b)$, there exists a $\bar{x} \in [a, b]$ such that $f(\bar{x}) = c$.

Solution: The argument works to show that $\begin{cases} f(\bar{x}) \leq c \\ f(\bar{x}) \not< c \end{cases} \implies f(\bar{x}) = c.$
We obtain the first relation via the continuity of f. Let

$$X = \{x \in [a, b] \mid f(x) \leq c\}$$

Since X is bounded ($X \subset [a, b]$) and non-empty ($a \in X$), there exists a supremum and $\sup(X) = \bar{x}$. Let us construct a sequence $(x)_n^\infty \to \bar{x} \subset X$ as $x_n \in X \cap [\bar{x} - 2^{-n}, \bar{x}]$. This sequence is well-defined because the intersection is never empty. Then, by the continuity of f, we have

$$\lim_{x\to\infty} f((x)_n^\infty) = f(\bar{x}) \leq c,$$

since $f(x_n) \leq c$.

We prove the second relation by contradiction. Let us assume that $f(\bar{x}) < c$. Let us choose $\varepsilon = c - f(\bar{x}) > 0$. Due to the continuity of f, the following applies:

$$|x - \bar{x}| < \delta \implies |f(x) - f(\bar{x})| < \varepsilon = c - f(\bar{x}).$$

Thus, $f(x) - f(\bar{x}) < \varepsilon = c - f(\bar{x}) \implies f(x) < c$. This means that we find function values in the interval $(\bar{x}, \bar{x}+\delta)$ for which $f(x) < c$. This contradicts the definition of $\bar{x}$, since it is the $\sup(X)$.

Thus, $f(\bar{x}) = c$ and the theorem follows.

Exercise 11
Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Prove that f is also uniformly continuous on $[a, b]$.

Solution: Assume, by contradiction, that f is not uniformly continuous on $[a, b]$. Then there

exists $\varepsilon > 0$ such that for every $\delta > 0$ one can find $x, y \in [a, b]$ with

$$|x - y| < \delta \quad \text{and} \quad |f(x) - f(y)| \geq \varepsilon.$$

Taking $\delta = 2^{-n}$ for each $n \in \mathbb{N}$, we obtain sequences $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ in $[a, b]$ with

$$|x_n - y_n| < 2^{-n} \quad \text{and} \quad |f(x_n) - f(y_n)| \geq \varepsilon.$$

Since $[a, b]$ is compact, the sequence $(x_n)_{n=0}^{\infty}$ has a convergent subsequence $(x_{n_k})_{k=0}^{\infty}$ with limit $\bar{x} \in [a, b]$. Then

$$|y_{n_k} - \bar{x}| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - \bar{x}| < 2^{-n_k} + |x_{n_k} - \bar{x}| \underset{k\to\infty}{\longrightarrow} 0,$$

so $y_{n_k} \to \bar{x}$ as well. By the continuity of f, the images of these sequences converge to $f(\bar{x})$:

$$\lim_{k\to\infty} f(x_{n_k}) = \lim_{k\to\infty} f(y_{n_k}) = f(\bar{x}).$$

Therefore

$$|f(x_{n_k}) - f(y_{n_k})| \leq |f(x_{n_k}) - f(\bar{x})| + |f(\bar{x}) - f(y_{n_k})| \underset{k\to\infty}{\longrightarrow} 0,$$

which contradicts our assumption. Hence, f is uniformly continuous on $[a, b]$.